Subject Description Form

Subject Code	ME29003/IC2105
Subject Title	Engineering Communication and Fundamentals
Credit Value	4 Training Credits
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject offers a wide spectrum of fundamental engineering practice that are essential for a professional engineer. This subject includes Engineering Drawing and CAD, Safety, Basic Mechatronic Practice, Mechanism Design Practice and Scientific Computing Languages that aims at providing fundamental and necessary technical skills to all year 1 students interested in engineering.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Describe the principles and conventional representation of engineering drawings according to engineering standards and be able to use it as a medium in technical communication and documentation with CAD application, modelling and practice with application in engineering; b) Interpret basic occupational health and industrial safety requirements for engineering practice; c) Explain common testing requirements; d) Apply scientific computing software for computing in science and engineering including visualization and programming. Upon completion of Stream A of the subject, students will be also able to: e) Design and implement simple mechatronics systems with programmable controller, software, actuation devices, sensing devices and mechanism. Upon completion of Stream B of the subject, students will be also able to: f) Design and fabricate simple mechanism assembly with standard components, fast prototyping processes and tolerance practices.

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Subject Synopsis/ Indicative Syllabus	1 (TM2009) Industrial Safety 1.1 Safety Management Overview, essential elements of safety management, safety training, accident management, and emergency procedures. 1.2 Safety Law F&IU Ordinance and principal regulations, OSH Ordinance and principal regulations. 1.3 Occupational Hygiene and Environmental Safety Noise hazard and control; dust hazard and control; ergonomics of manual handling. 1.4 Safety Technology Mechanical lifting, fire prevention, dangerous substances and chemical safety, machinery hazards and guarding, electrical safety, first aid, job safety analysis, fault tree analysis, and personal protective equipment. 2 (TM1340) Dimensioning and Tolerancing Practice 2.1 Measurement Principles of engineering drawing and orthographic projection; basic concept of dimensioning and tolerancing; introduction to common measuring tools and measurement practices such as steel rule, vernier calipers, micrometer, height gauge, optical projector and coordinate measuring machine (CMM). 2.2 Fitting Practice and Assembly Introduction to fasteners; introduction of hand tools and fitting practices such as filing, drilling, sawing, tapping and threading; assembly practice with fasteners and torque wrenches. 3 (TM8060) Computer Aided Design Fundamental 3.1 General concepts on CAD Parametric feature-based solid modelling; construction and detailing of solid features; solid model modification and its limitations. 3.2 Assembly modelling Bottom-up and top-down approaches for the generation of parts, subassemblies, and final assembly; mechanism design and its simulation methods. 3.3 Generation of engineering drawing Types of drawings including part drawing and assembly drawing; generation of 2D drawings from 3D parts and assemblies; drawing annotation.

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	One of the following as decided by hosting programme Stream A 4a (TM3019) MATLAB for Engineers and Scientists 4.1 Introduction of MATLAB Interactive calculations, random number generators, variables, vectors, matrices and string; mathematical operations, polynomial operation, data analysis and curve fitting, file I/O functions. 4.2 Problems solving with MATLAB Basic plotting, formatting graph, 2D and 3D plots, annotations, contour, mesh and surface plots, colormap. M-file programming & debugging; scripts, functions, logic operations, flow control and graphic user interfaces. 4.3 Basic data analysis and simulation Basic simulation using common functions for numerical integration and differentiations. Use of Simulink in the control of robotic systsms. Basic data analysis using common functions for filtering and spectral analysis. 5a (TM0512) Mechatronics Practice for Engineers 5.1 Overview of Mechatronics; Programmable Logic Controller (PLC); microcontroller systems; control systems (e.g. open loop, close loop, sequential) for monitoring and controlling mechatronic systems. 5.2 Sensor technologies used in mechatronic systems; computer vision and its application in robots/AI robots. 5.3 Typical mechanical power transmission systems; electromechanical devices (e.g. motors, solenoids); actuators with mechanical motion. Stream B 4b (TM3302) Python for Engineers and Scientists 4.1 Fundamental of Python Basic data type; variable and identifiers; constant, statement and expression, control structure and logic, string, tuple and list, set; object oriented concepts; interactive calculations and mathematical operations. 4.2 Problems solving with Python Functions and Python packages to solve engineering problem (e.g. plot displacement diagram). 4.3 Human Machine Interface (HMI)

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	Application development with data manipulation, visualisation and HMI by using data and graphics packages such as data processing, data plotting, visualisation, exploratory data analysis and graphic user interface. 5b (<u>TM1325) Fast prototyping for mechanism design</u> 5.1 Overview of basic mechanisms Basic principle of mechanical advantage mechanism; (e.g. gear, wheel and axle, linkages, pully, lever). 5.2 Fast prototyping techniques Basic principle and operations of 3D printing, FDM, SLA, DLP, pre-processing technique, part orientation, support structure, slicing, infill density, determination of different processing parameters for different applications, (e.g. light weight, heavily duty). laser machining & engraving operation techniques with its CAD preparation; basic 3D scanning operation; simulation of gear assembly and 4-bar linkage movement. 5.3 Performance evaluation of mechanism assembly Mechanism assembly by means of standard components, (e.g. gear, bolt and nuts, spacers) with Arduino motor control; performance evaluation; force and speed measurement; measurement of material properties.
Learning Methodology	The learning and teaching methods include lectures, workshop tutorials, and practical works. The lectures are aimed at providing students with an overall and concrete background knowledge required for understanding key issues in engineering communication, use of standard engineering components and systems, and importance of industrial safety. The workshop tutorials are aimed at enhancing students' in-depth knowledge and ability in applying the knowledge and skills to complete specific tasks. The practical works aim at facilitating students to review the diverse topics covered in this course and perform active learning with research, practice, questioning, and problem solving in a unified activity.

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Assessment Methods in Alignment with Intended Learning Outcomes

The assessment methods and weighting for Stream A and Stream B are same.

Assessment Methods	Remarks
1. Assignment	Individual in class hand-on practice assignment is designed to facilitate students to reflect and apply the knowledge periodically throughout the training.
2. Test	Test is designed to facilitate students to review the breadth and depth of their understanding on specific topics.
3. In-class learning logs	In-class learning log is designed to facilitate students to review their learning achievement and critize the outcomes by self-reflection.

Assessment Methods	Weighting (%)	Intended Learning Outcomes Assessed					
		a	b	c	d	e	f
Continuous Assessment							
1. Assignment/Project	77.5	✓	✓	✓	✓	✓	✓
2. Test	15		✓		✓		
3. In-class learning logs	7.5					*✓	*✓
Total	100						

Remark: ILO "e" and "f" will be assessed according to the Stream option, e.g. Stream A for "e" and Stream B for "f"

Student Study Effort Expected

Class Contact (Stream A)	**TM8060**	**TM2009**	**TM1340**	**TM0512**	**TM3019**
▪ Short lecture	7 Hrs.	7 Hrs.	3 Hrs.	7 Hrs.	7 Hrs.
▪ In-class Assignment/ Hands-on Practice	23 Hrs.	8 Hrs.	12 Hrs.	23 Hrs.	23 Hrs.

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	(Stream B)	TM8060	TM2009	TM1340	TM1325	TM3302
	• Short lecture	7 Hrs.	7 Hrs.	3 Hrs.	7 Hrs.	7 Hrs.
	• In-class Assignment/ Hands-on Practice	23 Hrs.	8 Hrs.	12 Hrs.	23 Hrs.	23 Hrs.
	Other Study Effort					
	▪ Nil					
	Total Study Effort					**120 Hrs.**

Reading List and References

Reference Software List:

1. AutoCAD from Autodesk Inc.
2. SolidWorks from Dassault Systèmes Solidworks Corp.
3. MATLAB from The Mathworks Inc.
4. Python from Python Software Foundation

Reference Standards and Handbooks:

1. BS EN ISO 128 – Technical product documentation. General principles of representation
2. Cecil H. Jensen, et al, Engineering Drawing and Design, McGraw-Hill, 2008.
3. IEEE Standard 315 / ANSI Y32.2 / CSA Z99 Graphic Symbols for Electrical and Electronics Diagrams.
4. IEC 61082 Preparation of Documents used in Electrotechnology.

Reference Books:
Training material, manual and articles published by Industrial Centre.